

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Political Engagement & Public Connect SaaS Platform

Project Type: Multi-Tenant White-Label SaaS

Platform: Responsive Web Application + Progressive Web App (PWA)

Primary Users: Political Leaders, Election Candidates, Political Organizations, Citizens, Supporters and Volunteers

Document Purpose: Development Team Handover

Version: 1.0

1. PROJECT OVERVIEW

1.1 Project Name

Working Name:

JanConnect / JanSampark

The final product name can be changed later without affecting the system architecture.

1.2 Project Objective

The objective is to develop a **multi-tenant, white-label Political Engagement SaaS Platform** that can be sold to different political leaders, election candidates and public representatives.

The platform will allow each client to have their own:

- Branded Web Application
- PWA-enabled mobile experience
- Admin Panel
- Public/Citizen Portal
- Area-wise data structure
- Registration form
- Content and promotional material
- Complaint management system
- Event system
- Poll system
- Membership system
- Poster/banner generator
- Volunteer management
- Communication tools
- Analytics dashboard

The same core platform must support multiple independent clients without mixing their data.

2. PRODUCT ARCHITECTURE

The complete system will consist of three major layers.

2.1 Super Admin Panel

Used by the SaaS platform owner/company.

Responsibilities:

- Create and manage clients/tenants
 - Configure subscription plans
 - Configure enabled modules
 - Manage white-label branding
 - Create client admin accounts
 - Manage domains/subdomains
 - Monitor platform usage
 - Manage system-level settings
 - View global analytics
 - Suspend or activate clients
-

2.2 Client / Leader Admin Panel

Each political leader or organization will have a separate admin environment.

The leader's authorized team can:

- Manage citizens
 - Manage members
 - Manage geographical areas
 - Publish banners and updates
 - Add development works
 - Manage complaints
 - Create events
 - Create polls
 - Manage gallery
 - Manage manifesto
 - Manage membership
 - Manage volunteers
 - Upload poster templates
 - Send notifications
 - View analytics
-

2.3 Public Web Application + PWA

The public-facing application will be responsive and work on:

- Mobile browsers
- Tablets
- Desktop
- Laptop

It will also support installation as a **Progressive Web App (PWA)**.

The user should be able to:

- Open the application through a URL
- Add/install it on the mobile home screen
- Launch it like an application
- Receive push notifications where supported
- Use a mobile-first interface

No separate Android or iOS application is required in Phase 1.

3. MULTI-TENANT SAAS ARCHITECTURE

Each political leader/client will be treated as a separate **Tenant**.

Example:

```
MASTER SAAS PLATFORM
├── Tenant A - Candidate / Leader A
│   ├── Admin Panel
│   └── Public Web App / PWA
├── Tenant B - Candidate / Leader B
│   ├── Admin Panel
│   └── Public Web App / PWA
└── Tenant C - Candidate / Leader C
    ├── Admin Panel
    └── Public Web App / PWA
```

Mandatory Requirement

Data belonging to one tenant must never be visible to another tenant.

Every tenant-specific record should be logically associated with:

tenant_id

This includes, but is not limited to:

- Users
 - Areas
 - Content
 - Complaints
 - Events
 - Polls
 - Memberships
 - Templates
 - Notifications
 - Analytics
-

4. USER ROLES

The system should use Role-Based Access Control (RBAC).

4.1 Platform-Level Roles

Super Admin

Full system access.

Permissions:

- Create tenants
- Edit tenants
- Suspend tenants
- Manage plans
- Manage subscriptions
- Manage global settings
- View all tenant statistics
- Manage support/admin accounts

Super Admin Staff

Permissions controlled by Super Admin.

Possible roles:

- Sales Manager
 - Support Executive
 - Technical Support
 - Finance Manager
-

5. TENANT-LEVEL ROLES

5.1 Leader / Owner

Highest-level user for a particular tenant.

Permissions:

- Full access to their tenant
 - Add/manage admins
 - View all reports
 - Configure available tenant settings
-

5.2 Tenant Admin

General administration access.

Can manage modules assigned by Leader/Super Admin.

5.3 Content Manager

Can manage:

- Banners
 - Blogs
 - News
 - Gallery
 - Videos
 - Development works
 - Manifesto
-

5.4 Complaint Manager

Can:

- View complaints
 - Update complaint status
 - Assign complaints
 - Add remarks
 - Upload resolution proof
-

5.5 Volunteer Manager

Can:

- Add volunteers
 - Assign areas
 - Create tasks
 - Monitor volunteer activity
-

5.6 Area Coordinator

Possible hierarchy:

District Coordinator
Block Coordinator
Gram Coordinator
Ward Coordinator

Area coordinators should only see data related to their assigned geographical area.

5.7 Public User

Public users can have categories:

Citizen
Supporter
Member
Volunteer

A single user may be upgraded from one category to another depending on tenant configuration.

6. ELECTION AND AREA CONFIGURATION

One of the most important requirements is that the platform must support different political/election levels.

Examples:

- Gram Pradhan
- BDC Member
- Block Pramukh
- Zila Panchayat Member
- MLA / Vidhan Sabha Candidate

- MP / Lok Sabha Candidate
- Other Political/Public Representative

The system must not hard-code one geographical hierarchy.

7. DYNAMIC AREA HIERARCHY

The system should support configurable hierarchical areas.

Example:

```

State
├── District
│   ├── Lok Sabha
│   │   ├── Vidhan Sabha
│   │   │   ├── Block
│   │   │   │   ├── Gram Panchayat
│   │   │   │   │   ├── Village
│   │   │   │   │   │   ├── Ward

```

However, not every client will require every level.

Example: Gram Pradhan

```

Gram Panchayat
├── Village
│   └── Ward

```

Example: MLA

```

Vidhan Sabha
├── Block
│   ├── Gram
│   │   └── Ward

```

Therefore, the hierarchy must be configurable per tenant.

Area Master Data Structure

Each area should contain:

```

Area ID
Tenant ID
Parent Area ID

```

Area Name
Area Type
Area Code
Status
Display Order

Suggested area types:

State
District
Lok Sabha
Vidhan Sabha
Block
Tehsil
Gram Panchayat
Village
Municipality
Zone
Ward
Booth
Custom Area

The system should support adding future area types.

8. TENANT ONBOARDING FLOW

When a new client purchases the SaaS product:

Step 1: Create Tenant

Required information:

Leader / Organization Name
Tenant Name
Contact Person
Mobile Number
Email
Election Type
Subscription Plan
Subscription Start Date
Subscription End Date

Step 2: Branding Setup

Platform Name
Logo
Leader Photo
Favicon
Primary Branding Settings
Secondary Branding Settings
Login Background
PWA Icon
Splash Screen

Step 3: Configure Domain

Options:

Subdomain
Custom Domain

Example:

xyz.platformdomain.com

or:

www.xyzleader.in

Step 4: Select Modules

Enable/disable:

About Leader
Development Works
Complaint System
Events
Polls
Membership
Photo Gallery
Video Gallery
Manifesto
Poster Generator
Volunteer Management
Notifications

Step 5: Configure Area Hierarchy

Select required levels.

Step 6: Configure Registration Form

Select which fields will appear.

Step 7: Create Admin Account

Admin receives credentials to access the tenant admin panel.

9. PUBLIC WEB APPLICATION

The public web application should have two levels of access.

9.1 Guest User

Before registration, users can view selected public information.

Pages/modules may include:

- Home
- About Leader
- Selected Development Works
- Latest Updates
- Events
- Photo Gallery
- Video Gallery
- Manifesto
- Social Media Links

Certain modules will require registration/login.

10. USER REGISTRATION SYSTEM

The registration system must be dynamic.

The tenant admin should be able to configure registration fields.

Default Fields

Full Name
Mobile Number
Profile Photo - Optional
Age / Date of Birth
Gender

Location Fields

Depending on tenant configuration:

Lok Sabha
Vidhan Sabha
District
Block
Gram Panchayat
Village
Ward
Booth

The area selection should be hierarchical.

Example:

Select Lok Sabha
↓
Available Vidhan Sabha
↓
Available Block
↓
Available Gram
↓
Available Ward

Dynamic Field Types

Admin should be able to create:

- Text
- Number
- Mobile
- Date
- Dropdown
- Multi-select
- Radio button

- Checkbox
- Area selector

Each field should support:

Field Name
Field Type
Required / Optional
Display Order
Options
Visibility

11. AUTHENTICATION SYSTEM

Preferred flow:

Mobile Number
↓
OTP Verification
↓
New User Registration
OR
Existing User Login

Requirements:

- OTP expiry
- Resend OTP controls
- Rate limiting
- Mobile number uniqueness per tenant
- Session management
- Logout functionality

Optional future authentication:

- Email login
- Password login
- Social login

12. PUBLIC APP HOME PAGE

The home page should be configurable.

Recommended sections:

Header
Leader Profile / Welcome
Main Promotional Banner
Quick Actions
Latest Updates
Leader's Works
Upcoming Events
Active Poll
Complaint CTA
Membership CTA
Poster Generator CTA
Gallery Preview
Social Media Links
Footer

Quick actions may include:

Our Works
Complaint
Events
Join Us
Poll
Create Poster

The admin should be able to control:

- Section visibility
- Display order
- Banner priority

13. ABOUT LEADER MODULE

The leader profile page should support:

- Profile Photo
- Cover Image
- Full Name
- Designation
- Short Introduction
- Full Biography
- Education
- Political Journey
- Social Work
- Achievements
- Awards
- Vision

- Mission

Optional:

- Timeline view
 - Video introduction
-

14. DEVELOPMENT WORKS MODULE

The system should allow the leader to publish development/social work activities.

Each work record should contain:

Title
Short Description
Full Description
Category
Area
Start Date
Completion Date
Status
Cover Image
Multiple Images
Video - Optional
Before Image - Optional
After Image - Optional

Categories

Default categories:

Road
Electricity
Water
Education
Healthcare
Employment
Women Empowerment
Youth Development
Infrastructure
Social Welfare
Other

Tenant admin should be able to create custom categories.

Work Status

Upcoming
In Progress
Completed

Users should be able to:

- View
- Filter by category
- Filter by area
- Share

Optional:

- Like
- Bookmark

15. COMPLAINT / JAN SAMASYA MODULE

This will be one of the primary modules.

Citizen Complaint Submission

User flow:

```
Select Complaint Category
      ↓
Select / Confirm Area
      ↓
Enter Complaint Title
      ↓
Enter Description
      ↓
Upload Images
      ↓
Optional Video
      ↓
Submit
```

The system should generate a unique complaint ID.

Example:

CMP-2026-000001

Complaint Fields

Complaint ID
User ID
Category
Title
Description
Area
Attachments
Created Date
Current Status
Assigned To
Priority
Remarks
Resolution Details

Complaint Status

Submitted
Under Review
Assigned
In Progress
Resolved
Closed
Rejected

Citizen Dashboard

Users should see:

Total Complaints
Pending
In Progress
Resolved
Closed

Users should be able to open a complaint and view:

- Timeline
- Status history
- Admin remarks
- Resolution proof

16. COMPLAINT MANAGEMENT – ADMIN

Admin should be able to:

- View all complaints
- Filter by area
- Filter by category
- Filter by date
- Filter by status
- Filter by assigned person
- Search by Complaint ID
- Assign complaint
- Change priority
- Add internal remarks
- Add public remarks
- Upload resolution proof
- Close complaint

Priority levels:

Low
Medium
High
Urgent

17. COMPLAINT ANALYTICS

Dashboard should display:

Total Complaints
Pending
In Progress
Resolved
Average Resolution Time

Additional analytics:

Complaints by Area
Complaints by Category
Complaints by Status
Monthly Complaint Trend
Top Problem Areas

18. EVENT MANAGEMENT MODULE

Admin can create events such as:

- Jan Sabha
- Rally
- Public Meeting
- Membership Campaign
- Press Conference
- Special Event
- Social Program

Each event:

Event Title
Banner
Description
Date
Start Time
End Time
Location
Map Link
Registration Required - Yes/No
Maximum Participants - Optional

Public users can:

- View event
- Register
- Mark Interested
- Mark Going
- Share event

Optional future functionality:

- QR-based event check-in
- Attendance reporting

19. POLL SYSTEM

Admin should be able to create polls.

Poll fields:

Question
Description - Optional

Options
Start Date
End Date
Target Audience
Area

Targeting options:

All Users
Specific Area
Members Only
Volunteers Only
Specific Age Group
Gender

Poll rules:

- One vote per authenticated user
- Poll result visibility configurable
- Start/end date validation

Admin analytics:

Total Votes
Votes by Area
Votes by Option
Participation Rate

20. MEMBERSHIP SYSTEM

The platform should support different membership types.

Example:

Citizen
Supporter
Volunteer
Active Member
Party Member

Membership should be configurable per tenant.

Membership process:

```
graph TD; A[User Registration] --> B[Membership Application]; B --> C[Optional Payment / Contribution]; C --> D[Admin Approval - Optional]; D --> E[Membership Activated];
```

Membership fields must be configurable.

21. PAYMENT / CONTRIBUTION MODULE

If membership requires payment or contribution:

```
graph TD; A[Membership Plan] --> B[Amount]; B --> C[Payment Status]; C --> D[Transaction Reference]; D --> E[Payment Date];
```

Payment statuses:

```
graph TD; A[Pending] --> B[Successful]; B --> C[Failed]; C --> D[Refunded];
```

Important requirement:

The payment module should be designed as an integration layer so different payment gateways can be connected.

The developer must ensure that the platform does not represent a payment as a legally compliant political donation merely by naming it “donation”; the applicable payment, organizational, tax, election and other compliance requirements must be validated before enabling this feature for a client.

22. DIGITAL MEMBERSHIP CARD

After membership approval, the system can generate a digital membership card.

Fields:

Member Photo
Member Name
Membership ID
Designation
Area
Issue Date
Expiry Date - Optional
QR Code
Tenant Branding

User actions:

- View card
- Download
- Share

Admin should be able to verify the member using Membership ID or QR code.

23. PHOTO GALLERY MODULE

Admin can create categories.

Example:

Public Meetings
Development Works
Campaign
Social Events
Press
Special Occasions

Each gallery item:

Title
Description
Category
Area - Optional
Date
Images

Public user actions:

- View
- Download, if enabled
- Share

Admin controls:

Allow Download
Allow Share
Public / Registered Users Only

24. VIDEO GALLERY MODULE

Videos can be added through:

- Direct upload
- External video URL/embed integration where supported

Video categories:

Speech
Interview
Development Work
Campaign
Public Message
Events
Other

User actions:

- Watch
- Share

Download should only be enabled where the content owner/admin explicitly permits it.

25. MANIFESTO MODULE

The leader/candidate can publish a manifesto.

Structure:

Manifesto Category
Title
Description
Cover Image
Supporting Documents - Optional

Categories:

Employment
Education
Healthcare
Farmers
Women
Youth
Infrastructure
Industry
Social Welfare
Other

Optional promise tracking:

Planned
In Progress
Completed

26. BLOG / NEWS / UPDATES MODULE

Admin can publish:

- News
- Announcements
- Articles
- Press Releases
- Leader Messages

Fields:

Title
Short Description
Full Content
Cover Image
Category
Author
Publish Date
Status

Status:

Draft
Scheduled

Published
Archived

Optional scheduling should allow content to publish automatically at a configured date/time.

27. POSTER / SOCIAL BANNER GENERATOR

This is a major engagement feature.

The tenant admin can upload pre-designed templates.

The public user can create personalized promotional banners.

User Flow

```
graph TD; A[Select Template] --> B[Upload / Select Profile Photo]; B --> C[Background Removal]; C --> D[Enter Name]; D --> E[Enter Designation]; E --> F[Optional Area]; F --> G[Generate Banner]; G --> H[Preview]; H --> I[Download]; I --> J[Share];
```

28. POSTER TEMPLATE MANAGEMENT

Admin should be able to:

- Upload template
- Define template category
- Enable/disable template
- Set expiry date
- Define editable fields

Example fields:

User Photo
Name
Designation
Area
Custom Text

Template categories:

Festival
Birthday
National Day
Political Campaign
Event Promotion
Congratulations
Support Campaign
General

29. BACKGROUND REMOVAL

When a user uploads a photo:

Original Image
↓
Background Removal Service
↓
Transparent Subject Image
↓
Placement on Template

Requirements:

- Image validation
- File size limit
- Background processing where required
- Error handling
- Temporary image storage policy
- User confirmation before final generation

The system should be designed so the background-removal provider can be changed later.

30. POSTER GENERATION

The poster engine should support:

- PNG
- JPG
- High-resolution output
- Mobile-friendly output
- Share-friendly dimensions

Recommended preset sizes:

1080 × 1080
1080 × 1350
1080 × 1920

The generated output should include tenant branding if configured.

31. NOTIFICATION SYSTEM

The system should support:

In-App Notifications

Examples:

- New event
- Complaint update
- Poll announcement
- New development work
- Membership update

Push Notifications

Where supported by the PWA/browser.

Admin should be able to target:

All Users
Members
Volunteers
Specific Area
Specific User Group

Each notification:

Title
Message
Image - Optional
Action URL - Optional
Target Audience
Schedule Date/Time
Status

32. SMS AND WHATSAPP INTEGRATION

The platform should be integration-ready for third-party providers.

Possible actions:

- OTP
- Membership confirmation
- Event reminder
- Complaint update
- Campaign communication

Because messaging providers and templates may involve additional charges and approval requirements, the SaaS system should track usage separately.

Usage:

SMS Sent
SMS Failed
WhatsApp Messages Sent
WhatsApp Messages Failed

The system should also allow the tenant to add external links such as:

- WhatsApp Channel
- WhatsApp Community/Group where appropriate
- Telegram
- Other social communities

33. SOCIAL MEDIA MANAGEMENT

Admin can configure links for:

- Facebook
- Instagram
- YouTube

- X
- WhatsApp
- Telegram
- Website

The links should be displayed in the public application.

34. VOLUNTEER MANAGEMENT

The system should support volunteer registration and management.

Volunteer fields:

Name
Mobile
Photo
Area
Assigned Role
Status
Joining Date

Roles may include:

General Volunteer
Ward Volunteer
Village Coordinator
Block Coordinator
Campaign Volunteer

Admin should be able to:

- Approve volunteer
 - Assign area
 - Assign tasks
 - Track activity
-

35. VOLUNTEER TASK MANAGEMENT

Admin can create tasks.

Example:

Task Title
Description

Assigned To
Area
Due Date
Priority
Attachments

Status:

Pending
Accepted
In Progress
Completed
Rejected

Volunteer can submit:

- Completion remark
- Images
- Report

36. USER PROFILE

Each public user should have:

Profile Photo
Name
Mobile
Age / DOB
Gender
Registered Areas
Membership Status
Volunteer Status
Complaint History
Event Registrations
Poll Participation
Generated Posters

Users should be able to edit only the fields allowed by tenant configuration.

37. MY AREA MODULE

Based on the user's registered area, the system should display personalized content.

Example:

My Lok Sabha
My Vidhan Sabha
My Block
My Gram
My Ward

Personalized content:

- Local works
- Local events
- Local announcements
- Area-specific polls
- Area-specific complaints where appropriate

38. PUBLIC USER DASHBOARD

After login, the user should see:

Welcome Section
Latest Updates
My Area
Quick Actions
Complaint Summary
Upcoming Events
Active Poll
Membership Status
Create Poster
Notifications

39. LEADER ADMIN DASHBOARD

Dashboard KPIs:

Total Registered Users
New Users Today
New Users This Week
Total Members
Total Volunteers

Total Complaints
Pending Complaints
In Progress
Resolved

Upcoming Events
Active Polls

Poster Generations
Active Users

Dashboard data should support date filters.

40. CITIZEN MANAGEMENT

Admin should have a searchable user management system.

Filters:

Name
Mobile
Registration Date
Area
Gender
Age Group
Membership Status
Volunteer Status

Admin actions:

- View user
- Edit permitted information
- Change status
- Add tag
- Export data subject to permissions
- Upgrade membership
- Assign volunteer role

41. USER TAGGING / CRM

Admin should be able to assign custom tags.

Examples:

Supporter
Volunteer
Youth
Student

Farmer
Business
Influencer
Media
Active Member

Tags can be used for filtering and communication segmentation.

42. CONTENT MANAGEMENT SYSTEM

The tenant admin should have centralized content management.

Content types:

Banner
Page
Blog
News
Development Work
Manifesto
Event
Poll
Gallery
Video

Common capabilities:

- Create
- Edit
- Delete
- Publish
- Unpublish
- Schedule
- Preview

43. BANNER MANAGEMENT

Admin can manage:

- Homepage banners
- Campaign banners
- Popup banners

Each banner:

Title
Image
Mobile Image - Optional
Action Link
Start Date
End Date
Priority
Status

44. ANALYTICS MODULE

User Analytics

Total Users
New Registrations
Active Users
Inactive Users
Registrations by Area
Registrations by Date

Demographic Analytics

Where such data has been collected and the tenant is permitted to process it:

Age Group
Gender Distribution
Area Distribution

Engagement Analytics

Most Viewed Content
Most Active Area
Most Participated Poll
Most Registered Event
Most Generated Poster

Complaint Analytics

Top Complaint Categories
Top Complaint Areas

Resolution Rate
Average Resolution Time

45. SUPER ADMIN PANEL

45.1 Dashboard

Global statistics:

Total Clients
Active Clients
Trial Clients
Expired Clients
Suspended Clients

Total Registered Users
Total Active Users
Total Storage Usage

45.2 Tenant Management

Super Admin should be able to:

- Create tenant
- Edit tenant
- Activate
- Suspend
- Renew subscription
- Change plan
- View tenant usage
- Impersonation/support access only with proper audit logging

46. SAAS SUBSCRIPTION MANAGEMENT

Subscription statuses:

Trial
Active
Expired
Suspended
Cancelled

Plans should support:

Plan Name
Monthly / Yearly Price
User Limit
Storage Limit
Enabled Modules
Support Level

47. FEATURE MANAGEMENT

Each tenant should have feature-level access control.

Example:

Feature	Tenant A	Tenant B
Complaint	Yes	Yes
Poll	Yes	No
Membership	Yes	Yes
Poster Generator	Yes	No
Volunteer Module	Yes	No

Feature flags must be checked at both:

- Frontend/UI level
- Backend/API authorization level

48. USAGE MANAGEMENT

The platform should track:

Registered Users
Storage Usage
SMS Usage
WhatsApp Usage
Notification Usage
Poster Generations

Usage limits can trigger:

Warning
Restriction
Overage Billing

depending on subscription configuration.

49. WHITE-LABEL CONFIGURATION

Each tenant should support:

Application Name
Logo
Favicon
PWA Icon
Leader Image
Theme Settings
Custom Domain
Footer Content
Privacy Policy URL/Page
Terms & Conditions URL/Page

The public interface must dynamically load the correct tenant branding based on domain/subdomain or configured tenant context.

50. PWA REQUIREMENTS

The public application must support Progressive Web App functionality.

Required

- Web App Manifest
- Installable experience where browser/platform supports it
- Responsive design
- App icon
- Splash/loading experience
- HTTPS
- Service Worker
- Offline fallback page
- Cache strategy

Optional Offline Features

Phase 2:

- Recently viewed content
- Basic offline page
- Saved profile information

The system should not allow stale cached data to incorrectly display sensitive or time-sensitive information.

51. RESPONSIVE DESIGN REQUIREMENTS

The application should be designed mobile-first.

Primary screen sizes:

Mobile
Tablet
Laptop
Desktop

Recommended mobile breakpoints should be decided by the development/design team, but all core functions must be usable without requiring horizontal scrolling.

52. API ARCHITECTURE

The backend should expose secure APIs.

Suggested groups:

Authentication API
Tenant API
Area API
User API
Content API
Complaint API
Event API
Poll API
Membership API
Payment API
Gallery API
Poster API
Notification API

Volunteer API
Analytics API

All protected APIs should validate:

Authentication
Tenant Access
Role
Permission
Area Access where applicable

53. DATABASE REQUIREMENTS

Suggested core entities:

```
tenants
tenant_domains
subscription_plans
subscriptions

users
roles
permissions
user_roles

areas
area_types
user_areas

dynamic_forms
form_fields
form_field_options
user_form_data

content_categories
banners
pages
blogs
news
development_works

complaints
complaint_categories
complaint_status_history
complaint_assignments
```

```
events
event_registrations

polls
poll_options
poll_votes

membership_plans
memberships
membership_cards
payments

galleries
gallery_items
videos

manifestos
manifesto_items

poster_templates
poster_generations

notifications
notification_logs

volunteers
volunteer_tasks
task_submissions

social_links

activity_logs
audit_logs
media_files
```

The final schema may change, but tenant isolation and scalable relationships are mandatory.

54. SECURITY REQUIREMENTS

The platform must implement:

- HTTPS
- Secure authentication
- OTP expiry
- Rate limiting
- Role-Based Access Control
- Tenant isolation
- Secure file upload validation
- Input validation

- SQL injection prevention
- XSS protection
- CSRF protection where applicable
- Secure password hashing if passwords are used
- Audit logging
- Admin activity tracking
- Backup strategy
- Error logging

Sensitive administrative actions should be logged.

Example:

```
User
Action
Timestamp
IP / Request metadata where lawfully collected
Affected Record
Old Value
New Value
```

55. DATA PRIVACY REQUIREMENTS

The platform will collect personal information such as:

- Name
- Mobile number
- Age or date of birth
- Gender if configured
- Area information

Therefore, the registration flow should support:

- Privacy Policy acceptance
- Terms acceptance
- Consent record
- Data access controls
- Account deletion/request workflow where required
- Configurable data retention

Sensitive personal data should not be exposed to unauthorized users.

56. FILE MANAGEMENT

Supported media:

JPG
JPEG
PNG
WEBP
MP4
Other approved formats

The system should enforce:

- Maximum file size
- File type validation
- Malware/security scanning strategy if available
- Optimized image generation
- Thumbnail generation
- CDN-ready storage architecture

57. SEARCH AND FILTERING

Search should be available for major data modules.

Examples:

Citizens

Name
Mobile
Area
Tag

Complaints

Complaint ID
User
Area
Category
Status

Content

Title
Category
Status
Date

58. EXPORT SYSTEM

Authorized admins may export selected data.

Supported formats:

```
CSV
Excel
```

Exports should be permission-controlled and logged.

Potential exports:

- Citizens
 - Members
 - Complaints
 - Events
 - Poll analytics
 - Volunteer reports
-

59. AUDIT LOG SYSTEM

The system should maintain an audit trail for important administrative actions.

Examples:

```
User Created
User Deleted
Role Changed
Complaint Updated
Content Published
Payment Status Updated
Subscription Changed
Tenant Suspended
```

Audit logs should generally not be editable by normal tenant admins.

60. ERROR HANDLING

The system should provide user-friendly messages.

Example:

Instead of:

500 Internal Server Error

Show:

Something went wrong. Please try again.

Technical errors should be logged server-side.

61. PERFORMANCE REQUIREMENTS

The application should be optimized for users with average mobile internet connections.

Requirements:

- Compressed images
- Lazy loading
- Pagination
- API pagination
- CDN-ready media
- Optimized database queries
- Caching where appropriate

Large datasets should never be loaded entirely on a single page.

62. RECOMMENDED TECHNOLOGY STACK

The final stack may be chosen by the development team, but the following architecture is recommended.

Frontend

Next.js / React
TypeScript

Backend

Node.js
NestJS or equivalent structured backend
TypeScript

Database

PostgreSQL

Cache / Queue

Redis

Storage

S3-compatible Object Storage

Deployment

Cloud Server
Docker-based deployment recommended
CI/CD
Environment separation

Environments:

Development
Staging
Production

63. DEVELOPMENT PHASES

PHASE 1 – CORE MVP

Super Admin

- Tenant management
- Plan management
- Subscription management
- Branding
- Feature control

Tenant Admin

- Dashboard
- Area management

- User management
- Content management
- Development works
- Complaint management
- Banner management

Public Application

- Landing pages
- Registration
- OTP login
- Dynamic registration form
- Area selection
- Home dashboard
- Works
- Complaint
- Gallery
- Events
- User profile

PWA

- Installable web application
 - Mobile responsive
 - Manifest
 - Service worker
 - Offline fallback
-

64. PHASE 2 – ENGAGEMENT

Add:

- Polls
 - Membership
 - Payment integration
 - Digital membership card
 - Notifications
 - Volunteer management
 - CRM tags
 - Advanced analytics
-

65. PHASE 3 – POSTER AND VIRAL ENGAGEMENT

Add:

- Poster templates
- User photo upload

- Background removal
 - Personalized text
 - Poster generation
 - Download
 - Sharing
-

66. PHASE 4 – ADVANCED CAMPAIGN MANAGEMENT

Potential future features:

- Campaign tasks
 - Door-to-door campaign reporting
 - Coordinator hierarchy
 - Booth-level management
 - Advanced area analytics
 - Advanced communication
 - AI-assisted insights
 - Lead scoring/engagement scoring, subject to applicable legal and policy requirements
-

67. DEVELOPMENT PRIORITY

The development team should prioritize the architecture before visual features.

Priority 1

Multi-Tenant Architecture
Tenant Isolation
RBAC
Dynamic Area Hierarchy
Dynamic Registration Form

Priority 2

Public Registration
User Management
Content Management
Development Works
Complaint System

Priority 3

PWA
Notifications
Events
Gallery
Analytics

Priority 4

Membership
Payments
Volunteer System
Polls

Priority 5

Poster Generator
Background Removal
Advanced Campaign Features

68. UI/UX REQUIREMENTS

The UI should be:

- Clean
- Modern
- Mobile-first
- Easy for non-technical political staff
- Hindi + English ready
- Fast to navigate
- Minimal click flow

Admin Panel

Recommended:

Left Sidebar
Top Header
Dashboard Cards
Data Tables
Filters

Charts
Responsive Layout

Public Application

Recommended:

Mobile-first layout
Large touch targets
Simple navigation
Card-based sections
Clear CTA buttons
Minimal form complexity

69. LANGUAGE SUPPORT

Phase 1 should support:

English
Hindi

The architecture should support adding additional languages later.

All user-facing text should preferably be localization-ready rather than hard-coded throughout the frontend.

70. SYSTEM WORKFLOW SUMMARY

New Client Setup

```
Super Admin Login
  ↓
Create Tenant
  ↓
Select Subscription Plan
  ↓
Select Election / Representative Type
  ↓
Configure Area Hierarchy
  ↓
Configure Modules
  ↓
```


Upload Branding
↓
Create Tenant Admin
↓
Configure Domain
↓
Publish Platform

Public User Flow

Open Website/PWA
↓
Explore Public Content
↓
Register / Login
↓
Mobile OTP Verification
↓
Complete Dynamic Form
↓
Select Area
↓
Enter Dashboard
↓
Access Available Modules

Complaint Flow

Citizen Creates Complaint
↓
System Generates Complaint ID
↓
Complaint Submitted
↓
Admin Reviews
↓
Assigns Responsible Person
↓
Status Updated
↓
Work in Progress
↓
Resolution Proof Uploaded
↓
Complaint Resolved / Closed

↓
Citizen Receives Update

71. NON-FUNCTIONAL REQUIREMENTS

Scalability

The system should support:

- Multiple tenants
- Large numbers of users
- Large media libraries
- High concurrent traffic during campaign periods

Availability

Production architecture should support:

- Automated backups
- Monitoring
- Error logging
- Recovery procedures

Maintainability

Code should follow:

- Modular architecture
- Proper documentation
- API documentation
- Environment configuration
- Version control
- Migration management

72. DEVELOPER DELIVERABLES

The development team should provide:

Source Code

Complete source code for:

Frontend
Backend

Admin Panel
Public Application
PWA Configuration

Documentation

Required documentation:

- Setup Guide
- Environment Variables Guide
- Database Schema/ERD
- API Documentation
- Deployment Guide
- Admin/User Role Documentation

Infrastructure

Document:

- Hosting requirements
- Database requirements
- Storage configuration
- Background worker/queue requirements
- Third-party services

73. THIRD-PARTY SERVICES TO BE CONFIGURED

The system should use abstraction/integration layers for:

OTP Provider
SMS Provider
WhatsApp Provider
Payment Gateway
Push Notification Service
Cloud Storage
Background Removal API
Analytics Service

Provider credentials should not be hard-coded in source code.

74. ACCEPTANCE CRITERIA

The Phase 1 system will be considered functionally complete when:

1. Super Admin can create a new tenant.
 2. Super Admin can configure tenant branding.
 3. Super Admin can enable/disable modules.
 4. Tenant can have an independent admin account.
 5. Tenant data is isolated from other tenants.
 6. Tenant can configure geographical areas.
 7. Tenant can configure registration fields.
 8. Public users can register through OTP.
 9. Users can select hierarchical areas.
 10. Tenant admin can manage users.
 11. Tenant admin can publish banners and content.
 12. Tenant admin can add development works.
 13. Users can submit complaints.
 14. Admin can manage complaint status.
 15. Users can track their complaints.
 16. Events can be created and viewed.
 17. Gallery can be managed.
 18. The public application is fully responsive.
 19. The application can be installed as a PWA where supported.
 20. Role and permission restrictions work correctly.
 21. Unauthorized users cannot access protected tenant data.
 22. Major actions are logged appropriately.
-

75. FUTURE EXPANSION POSSIBILITIES

The architecture should allow future modules such as:

- Booth Management
 - Campaign Management
 - Volunteer Attendance
 - QR-based Event Attendance
 - Survey System
 - Feedback System
 - AI Chatbot
 - AI Content Assistant
 - Speech/Video Transcription
 - Advanced Political CRM
 - Call Center Integration
 - Lead Management
 - Task Automation
 - Election Campaign Dashboard
 - Multi-organization management
-

76. FINAL PRODUCT DEFINITION

The final product should be understood as:

A Multi-Tenant White-Label Political Engagement and Public Relationship Management SaaS Platform delivered through a Responsive Web Application and Progressive Web App.

The platform should allow the SaaS owner to onboard multiple political leaders or organizations and provide each one with a separately branded, configurable public engagement platform.

The core design principles are:

```
One Core Platform
      +
Multiple Independent Clients
      +
Dynamic Area Hierarchy
      +
Configurable Registration
      +
Role-Based Access
      +
Feature Toggles
      +
White-Label Branding
      +
Responsive Web App
      +
PWA Experience
```

The platform must be designed in a modular way so that new clients can be onboarded primarily through configuration rather than requiring separate software development for every political leader.

77. IMPORTANT DEVELOPMENT NOTE

The development team should **not build this as a collection of fixed pages for one political leader**.

The core product must be developed as a configurable SaaS engine.

The following four components are foundational and must be architecturally correct from the beginning:

1. Multi-Tenant Data Isolation

Every client must have isolated data.

2. Dynamic Geographic Hierarchy

The system must support different political/election structures without code changes for every client.

3. Dynamic Forms

Registration and selected workflows must be configurable per tenant.

4. Feature-Based SaaS Configuration

Each client must be able to receive a different set of enabled modules.

If these four foundations are implemented correctly, the platform can later scale from a local Gram Pradhan candidate to an MLA, MP, political organization or other public representative without rebuilding the entire application.